

Automated Network Request Management in ServiceNow

ServiceNow Custom Application

Project	Automated Network Request Management
Platform	ServiceNow
Interface	Service Portal
WorkFlow	Catalog → Records → Approval → Task → Status → Notification

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1. Project Overview

The **Automated Network Request Management in ServiceNow** project provides a structured way to submit and manage network-related service requests. It automates the process from request submission to approval, Network Task creation, status updates, and notifications.

The project uses **Service Catalog, Flow Designer, custom tables, Approvals, Business Rules, ACLs, and Notifications** to manage the complete request lifecycle.

The solution provides the following capabilities:

- Users can submit network requests through the Service Portal.
- Required request information is captured using structured catalog variables.
- The Requested For field is automatically populated with the current user.
- ServiceNow creates Request and Requested Item records.
- Flow Designer creates the Network Database record and routes the request for approval.
- An approved request generates a Network Task for the fulfilment team.
- Status changes and email notifications keep concerned users informed.

2. Problem Statement and Objectives

Network service requests can become difficult to manage when they are submitted through unstructured channels. This project provides a controlled ServiceNow process for collecting request details, obtaining approval, creating fulfilment tasks, and communicating progress.

Objectives

- Create a centralized Network Request Catalog Item.
- Capture request details using structured catalog variables.
- Automate approval and Network Task creation.
- Track request status throughout the lifecycle.
- Send notifications to concerned users.

Improve request management through Flow Designer, ACLs, and Update Sets.

3. Application and Data Architecture

The project is developed on the ServiceNow platform using the Service Catalog, custom tables, Flow Designer, and Service Portal.

The application follows this workflow:

Catalog Item → Request → Requested Item → Network Database → Approval
→ Network Task → Status Update → Notification

The main components are:

- Service Catalog: Allows users to submit network requests.
- Catalog Variables: Collect required request details.
- Custom Tables: Store Network Database and Network Task records.
- Flow Designer: Automates approvals and task creation.
- Service Portal: Provides the user interface for submitting and tracking requests.
- Notifications: Inform users about approval and request progress.

The screenshot displays the ServiceNow 'Network Request' form within the Service Catalog. The breadcrumb trail at the top indicates the path: Home > Service Catalog > Network > Network Request. The form is titled 'Network Request' and includes a 'Network Request Management' section. It features several input fields for 'Requester Information', including 'Email ID' (with a 'Proof of document' upload button), 'Opened on behalf of' (a dropdown menu), 'Phone Number', and 'type of device' (a dropdown menu with 'Laptop' selected). A 'Username' field is also present. On the right side, there is a 'Quantity' field set to 1, a 'Delivery Time' of 0 Days, and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. Below these, a 'Required information' section lists 'Email ID', 'Phone Number', and 'Address' as required fields, with a 'Device Details' link. The top navigation bar includes links for Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile for 'System Administrator'. A search bar labeled 'Search Catalog' is located in the top right corner of the form area.

4. Requirement Analysis and Planning

The project requirements were identified to automate the complete network request process.

Functional Requirements:

- Users should submit network requests through the Service Portal.
- Required details must be captured using catalog variables.
- The system should create request and requested item records.
- Approval should be automated using Flow Designer.
- Approved requests should create Network Task records.
- Users should receive status updates and notifications.

Planning:

The implementation was divided into catalog configuration, table creation, flow automation, approval setup, notifications, testing, and deployment.

5. Catalog Item and Variable Configuration

A Network Request Catalog Item was created to allow users to submit network-related requests.

The catalog item contains structured variables to collect the required request information. The Requested For field is automatically populated with the current user.

These variables ensure that complete and accurate details are captured before the request is submitted.

The screenshot shows the 'Catalog Item' configuration page for 'Network Request'. The top navigation bar includes 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue banner at the top states: 'Build and modify items faster with the improved Catalog Builder. Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. Enter a Name and Short description to display for the item. Enter a Price, approvals, variables, and other information as needed.'

The configuration fields are as follows:

- Name:** Network Request
- Catalogs:** Service Catalog
- Category:** Network Standard Changes
- State:** -- None --
- Checked out:** -- None --
- Owner:** System Administrator
- Secondary owners:** (empty)
- Application:** Global
- Active:** ☒
- Fulfillment automation level:** Unspecified

The bottom navigation bar includes 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Short description' field is populated with 'Network Service Request'.

Fig 5.1 Catalog Item Created

The screenshot shows the 'Variable Set' configuration page for 'Requester Information'. The top navigation bar includes 'Update' and 'Delete'. The configuration fields are as follows:

- Title:** Requester Information
- Internal name:** requester_information
- Order:** 100
- Type:** Single Row
- Application:** Global
- Display title:** ☒
- Layout:** 2 Columns Wide, one side, then the other

The bottom section shows a table of variables:

Name	Type	Question	Order
email_id	Single Line Text	Email ID	
opened_on_behalf_of	Reference	Opened on behalf of	
phone_number	Single Line Text	Phone Number	
proof_of_document	Attachment	Proof of document	
username	Single Line Text	Username	

The bottom right corner shows '1 to 5 of 5'.

Fig 5.2 Variable Configuration

6. Service Portal Interface and Usability

The **Service Portal** provides a simple interface for users to submit network requests.

Users can open the Network Request Catalog Item, enter the required details, and submit the request. The structured form helps users provide complete information and makes the request process easier to follow.

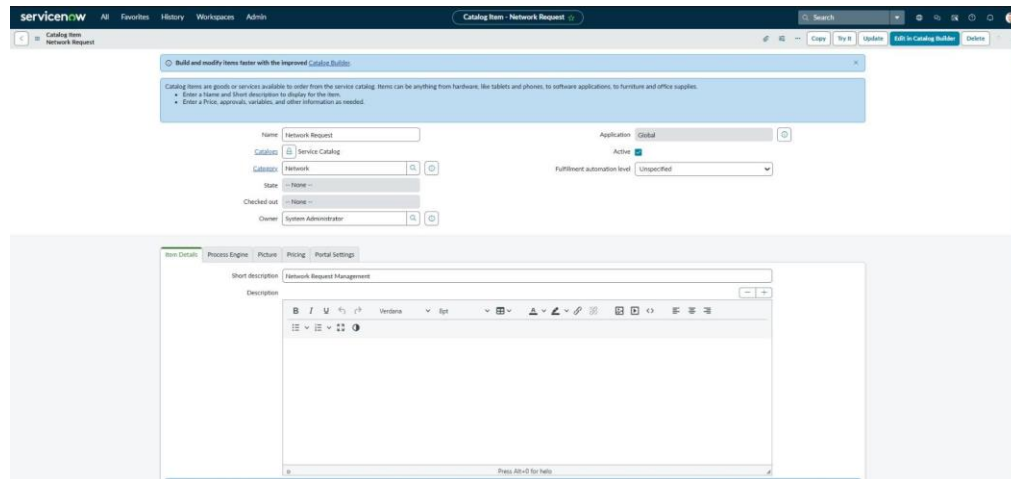
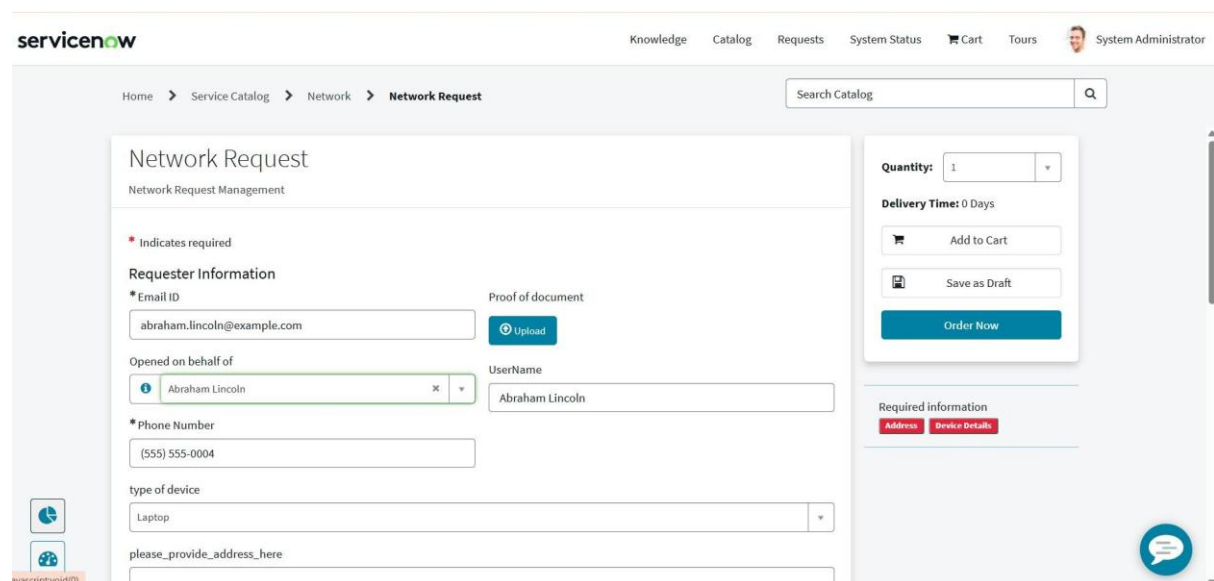


Fig 6.1 Creation of Service Catalog



7. Flow Designer Automation

Flow Designer automates the network request process after a user submits the catalog item.

The flow creates the required Network Database record, routes the request for approval, and creates a Network Task when the request is approved. It also supports status updates and notifications.

Workflow:

Request Submission → Network Database Record → Approval → Network Task Creation → Status Update → Notification

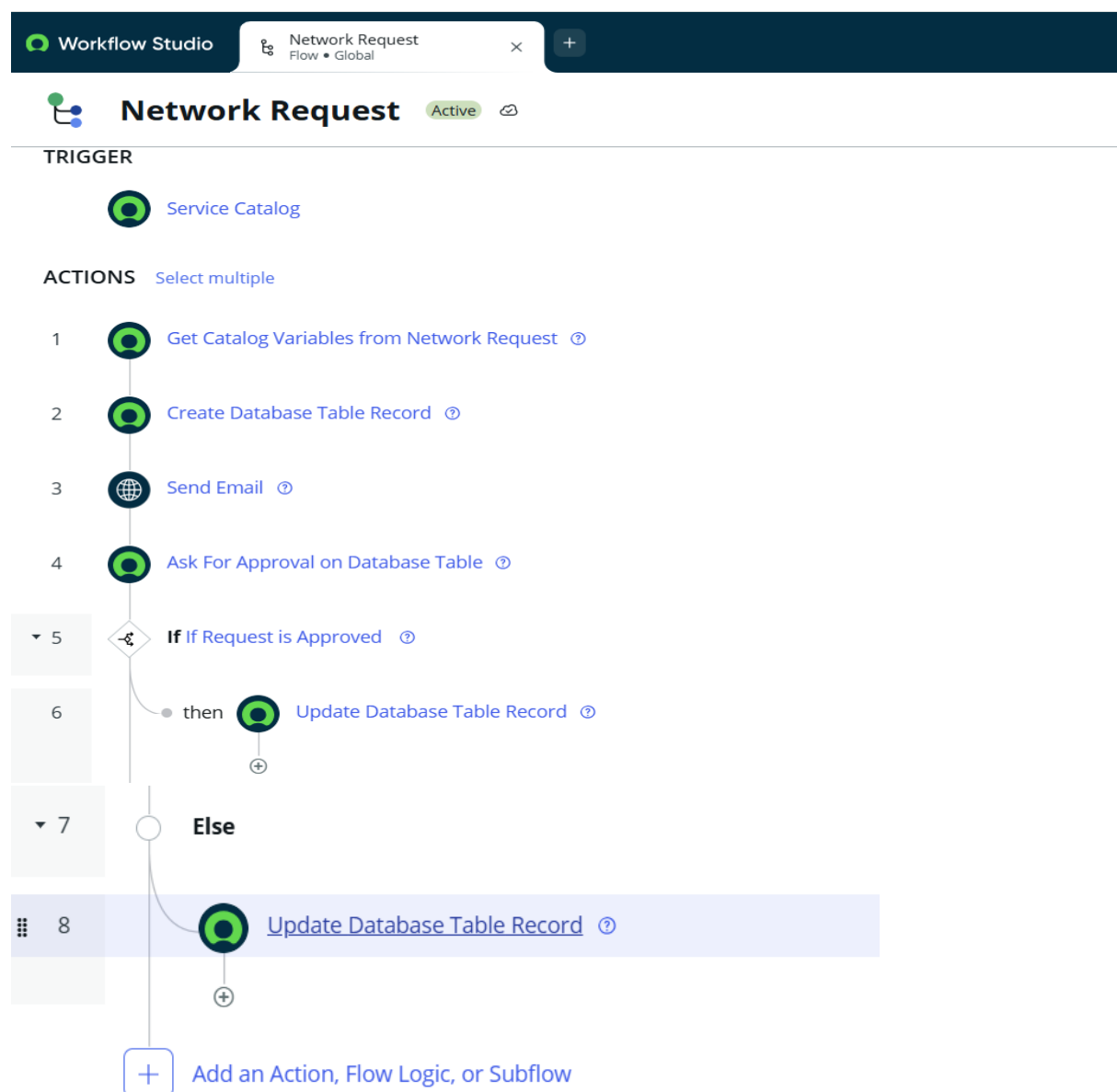
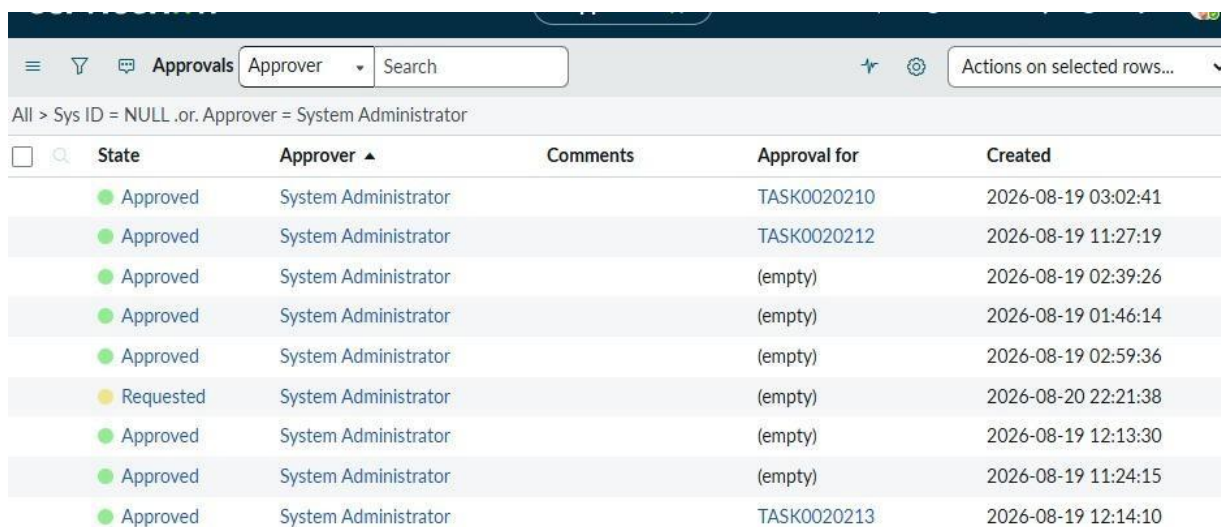


Fig 7.1 Creation of Flow

8. Approvals and Network Task

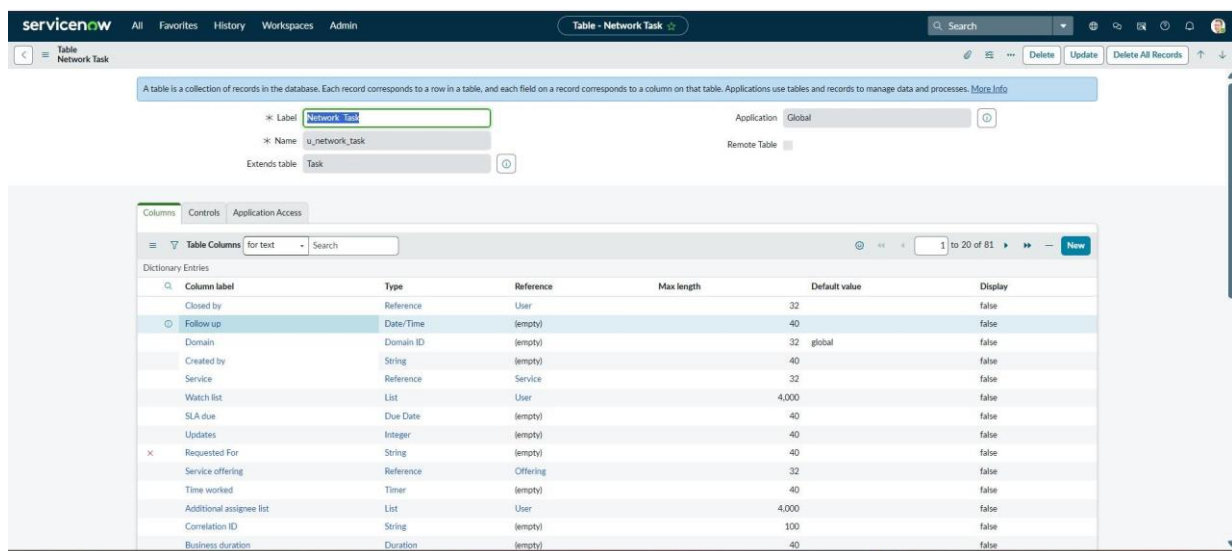
After the Network Database record is created, the request is sent for approval.

- The approver reviews the submitted request.
- If approved, a **Network Task** is created for fulfilment.
- The task helps the concerned team manage and complete the network request.
- The request status is updated throughout the process.



	State	Approver	Comments	Approval for	Created
☐	Approved	System Administrator		TASK0020210	2026-08-19 03:02:41
	Approved	System Administrator		TASK0020212	2026-08-19 11:27:19
	Approved	System Administrator		(empty)	2026-08-19 02:39:26
	Approved	System Administrator		(empty)	2026-08-19 01:46:14
	Approved	System Administrator		(empty)	2026-08-19 02:59:36
	Requested	System Administrator		(empty)	2026-08-20 22:21:38
	Approved	System Administrator		(empty)	2026-08-19 12:13:30
	Approved	System Administrator		(empty)	2026-08-19 11:24:15
	Approved	System Administrator		TASK0020213	2026-08-19 12:14:10

Fig 8.1 – Approval state evidence used for data-integrity validation



Column label	Type	Reference	Max length	Default value	Display
Closed by	Reference	User	32		false
Follow up	Date/Time	(empty)	40		false
Domain	Domain ID	(empty)	32	global	false
Created by	String	(empty)	40		false
Service	Reference	Service	32		false
Watch list	List	User	4,000		false
SLA due	Due Date	(empty)	40		false
Updates	Integer	(empty)	40		false
Requested For	String	(empty)	40		false
Service offering	Reference	Offering	32		false
Time worked	Timer	(empty)	40		false
Additional assignee list	List	User	4,000		false
Correlation ID	String	(empty)	100		false
Business duration	Duration	(empty)	40		false

Fig 8.2 Network Task

9. Business Rules, UI Logic and ACLs

Business Rules and UI logic support the proper functioning of the network request process.

- **Business Rules:** Help manage request and task-related record processing.
- **UI Logic:** Supports the required form behaviour.
- **Access Control Lists (ACLs):** Control access to application records and ensure that only authorized users can view or modify them.

These configurations improve data security and maintain the correct request workflow.

Related Links

Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
Types of devices is others	empty		true	true	false	2026-08-21 03:29:12	100

9.1 UI Policy

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_network_database	Allow if	create	record	true	admin	2026-08-18 07:39:40
u_network_database	Allow if	write	record	true	admin	2026-08-18 07:39:41
u_network_database	Allow if	read	record	true	admin	2026-08-18 07:39:40
u_network_database	Allow if	delete	record	true	admin	2026-08-18 07:39:41

9.2 ACL created

10. Notifications and Status Tracking

The system updates the request status during the fulfilment process.

Notifications inform users about important request updates, such as approval and completion. This helps users track their network requests and understand the current progress.

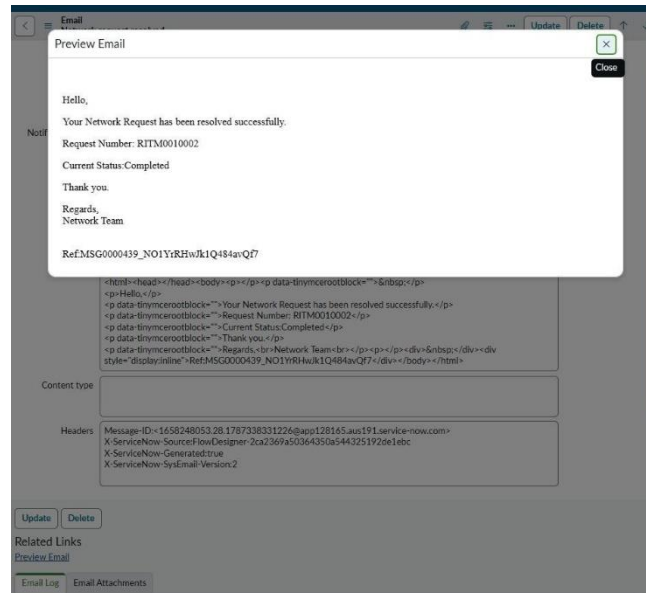


Fig 10.1 Email notification output

Home > Request Summary - REQ0010002

Search Catalog

Thank you!
Your request has been submitted.

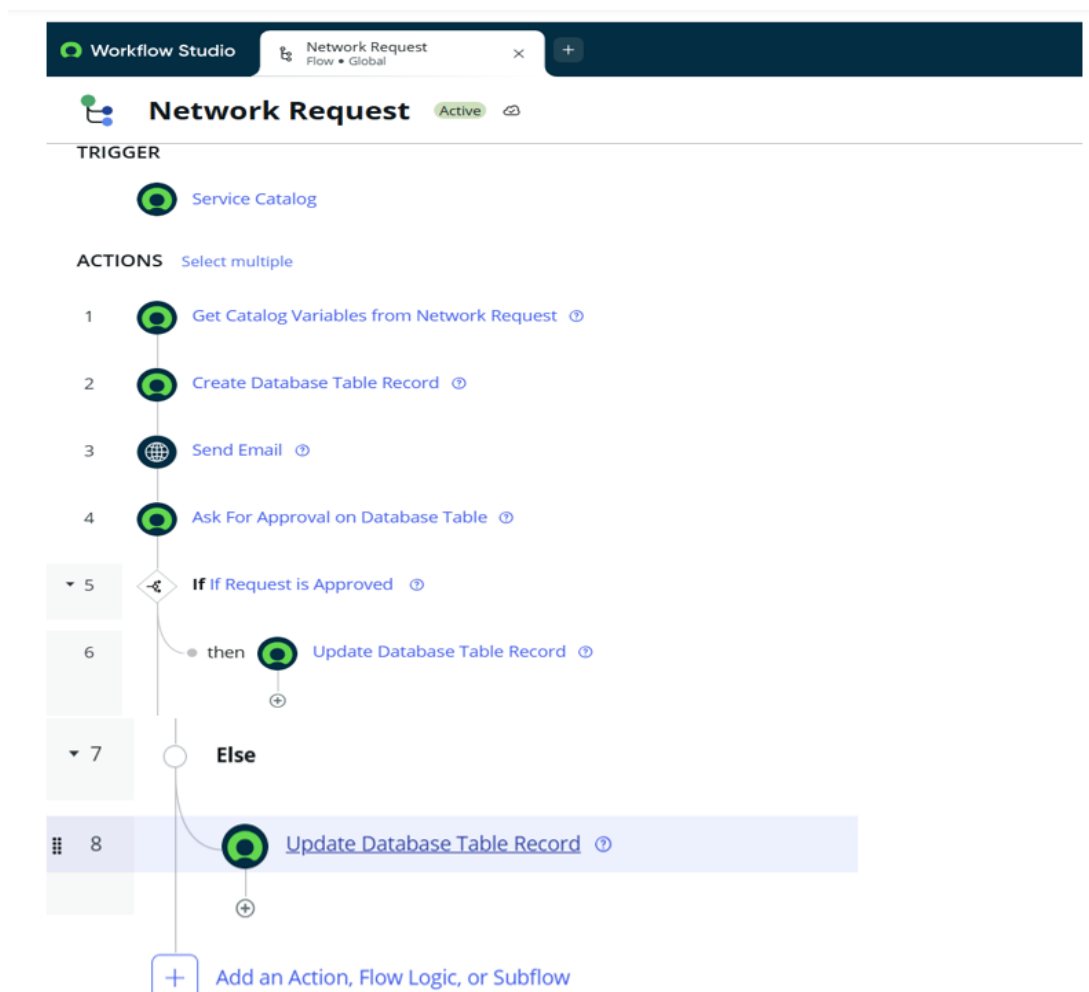
Submitted : 2026-08-20 22:21:33
Request Number : **REQ0010002**
Estimated Delivery : 2026-08-20

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Network Request	2026-08-20		---	1	---

11. End-to-End Functional Flow

The complete network request process follows these steps:

1. The user submits a Network Request through the Service Portal.
2. ServiceNow creates the Request and Requested Item records.
3. Flow Designer creates the Network Database record.
4. The request is sent for approval.
5. After approval, a Network Task is created.
6. The concerned team works on the task.
7. The request status is updated during fulfilment.
8. Notifications inform the user about important updates and completion.

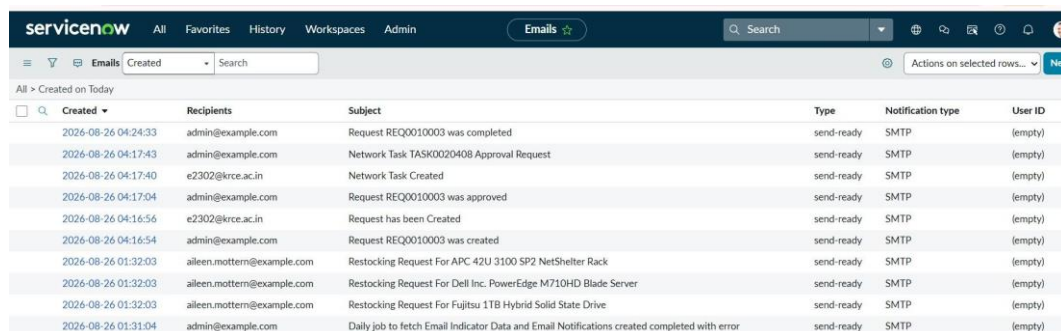


12. Testing and Validation

The project was tested to verify that the complete network request process works correctly.

The following functions were validated:

- Network Request Catalog Item submission.
- Mandatory field validation.
- Automatic population of the Requested For field.
- Creation of Request and Requested Item records.
- Creation of the Network Database record.
- Approval process.
- Network Task creation after approval.
- Request status updates.
- Email notifications.
- Access control validation.



The screenshot shows the ServiceNow interface with the 'Emails' list view selected. The table displays a list of email notifications with columns for Created, Recipients, Subject, Type, Notification type, and User ID. The data is filtered by 'Created' and shows 10 rows of notifications.

Created	Recipients	Subject	Type	Notification type	User ID
2026-08-26 04:24:33	admin@example.com	Request REQ0010003 was completed	send-ready	SMTP	(empty)
2026-08-26 04:17:43	admin@example.com	Network Task TASK0020408 Approval Request	send-ready	SMTP	(empty)
2026-08-26 04:17:40	e2302@krce.ac.in	Network Task Created	send-ready	SMTP	(empty)
2026-08-26 04:17:04	admin@example.com	Request REQ0010003 was approved	send-ready	SMTP	(empty)
2026-08-26 04:16:56	e2302@krce.ac.in	Request has been Created	send-ready	SMTP	(empty)
2026-08-26 04:16:54	admin@example.com	Request REQ0010003 was created	send-ready	SMTP	(empty)
2026-08-26 01:32:03	alileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-08-26 01:32:03	alileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-08-26 01:32:03	alileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-08-26 01:31:04	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)

13. Deployment, Documentation and Presentation

The project was configured and tested on the ServiceNow platform. The implementation includes the catalog item, custom tables, Flow Designer automation, approvals, Network Tasks, notifications, ACLs, and Update Sets.

The project documentation explains the configuration steps, workflow, testing results, and overall functionality. Screenshots and execution evidence were used to demonstrate the working process.

14. Conclusion and Future Enhancements

The Automated Network Request Management system centralizes network request submission and automates approval, task creation, status tracking, and notifications.

The project reduces manual effort and improves visibility throughout the request lifecycle.

Future Enhancements:

- Add more request types and approval levels.
- Introduce additional fulfilment automation.
- Create dashboards and detailed reports.
- Expand role-based access and validation.
- Improve portal tracking and confirmation pages.